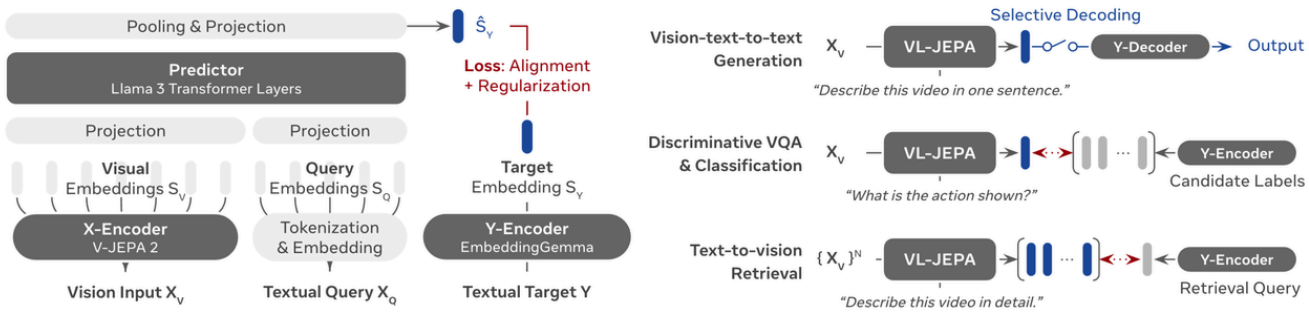
VL-JEPA: Joint Embedding Predictive Architecture for Vision-language

> #WorldModel #JEPA #HybridArchitecture #CV



- What problems does this architecture solve?
  - The current VLMs are hard to train. They not only need to learn the core semantics but also need to learn the task-irrelevant language features. In the token space, sentences with a similar meaning but different expressions can be orthogonal, making the convergence hard.
  - Due to the autoregressive generation, the latency of traditional VLMs can be very high. The semantics cannot be fully shown unless the autoregressive generation is done, so it is not fit for the streaming or long-horizon videos.
- How does this architecture work?
  - Instead of explicitly generating the tokens autoregressively, based on the philosophy of JEPA, the model will do inference and prediction in the latent space.
  - The task can be represented as a triplet:

    $< X_V, X_Q, Y >$, where:
    - X_V is the visual input.
    - X_Q is the NL query.
    - Y is the textual target.
  - There are four main structures in VL-JEPA:
    - X-encoder: Visual input to visual embedding. $X_V \to S_V$
    - Predictor: Based on the NL query, visual embedding to generate target embedding. $X_V, X_Q \to \hat{S}_Y$
    - Y-encoder: Groundtruth NL to groundtruth target embedding. $Y \to S_Y$
    - Y-decoder: Translate the predicted target embedding to NL $\hat{S}_Y \to \hat{Y}$. *This part will not be in the main training process.*
  - 2-Step Training:
    - Objective: InfoNCE Loss. Use the groundtruth label as the positive example and the rest of the labels as the negative labels to run:

      $$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(\text{sim}(q,k_+)/\tau)}{\exp(\text{sim}(q,k_+)/\tau)+\sum_{i=1}^{K}\exp(\text{sim}(q,k_i)/\tau)}$$

      1. Pretraining Stage: The dataset contains scenes and its corresponding text description.

During this stage, the model is not taught to answer the questions. Instead, only visual input will be fed to the model and the model will output the latent representation to describe the scene it sees.

2. SFT Stage: Both visual info and user query will be fed to the model to teach it to answer the query from the user.

- Selective Decoding
  - As mentioned in the problems section, the decoding process is expensive. VL-JEPA has no autoregressive process, so it can represent the complete semantic meaning without decoding.
  - When will it decode? There will be a variance monitor. If it finds that the latent representation appears to have a large variance (drastic semantic change), the Y-decoder will be triggered to translate it to NL.
  - This saves the computes, lowers the latency, and makes it better fit for long-horizon videos.

- Experiment Results
  - Higher training efficiency: In a fair comparison with all variables held constant (same visual encoder, same data, same number of training steps), VL-JEPA, which performs embedding-based prediction, achieves superior performance using only half the trainable parameters of a traditional VLM (approximately 0.5B vs. 1B).
  - Excellent inference efficiency: Thanks to its “selective decoding” mechanism, VL-JEPA reduces the number of computationally expensive text decoding operations by approximately 2.85 times when processing video streams, while maintaining equivalent output quality (CIDEr score).
  - Robust general-purpose multi-task capabilities: Without any architectural modifications, its single embedding space naturally supports zero-shot classification, video retrieval, and discriminative VQA tasks. As a lightweight model with only 1.6 billion parameters, it outperforms CLIP and SigLIP2 on average across multiple video classification and retrieval tasks, and rivals much larger traditional VLMs (such as InstructBLIP and Qwen-VL) on VQA tasks.

- Work Cited
  - Paper
  - Chat History
  - NotebookLM